Lab Report: Text classification with neural networks

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 80 percent training data and the rest for testing. Each experiment was repeated 3 times with different random seeds.

2 Results

Figure 1 shows the accuracy by dataset.

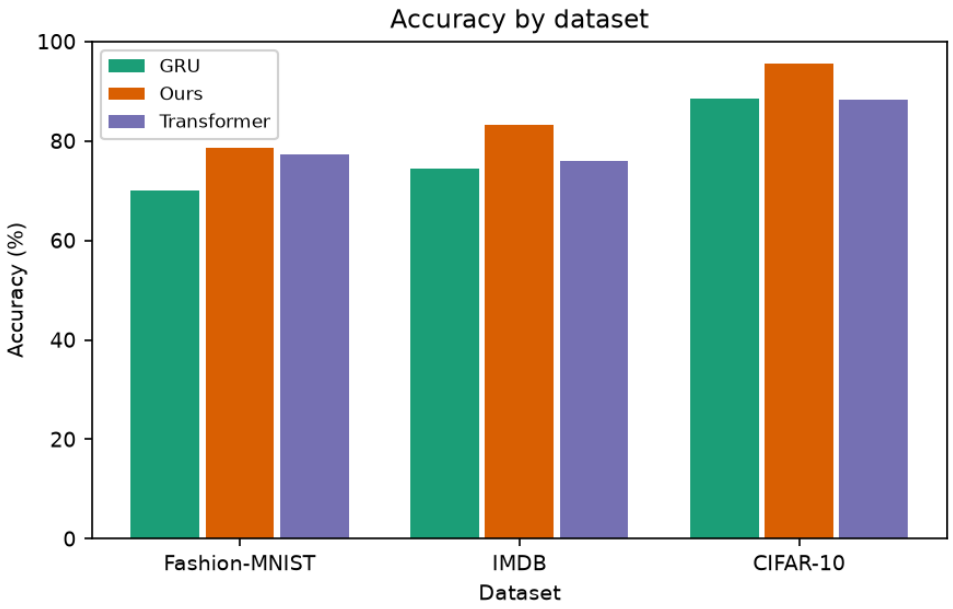


Figure 1: Accuracy by dataset for the models in this experiment.

As shown in Figure 1, on CIFAR-10, GRU obtains a accuracy of 88.5%. Ours achieves 75.5% accuracy on CIFAR-10. Ours outperforms Transformer by 7 percentage points on CIFAR-10.

Figure 2 shows the accuracy of MobileNet by dataset.

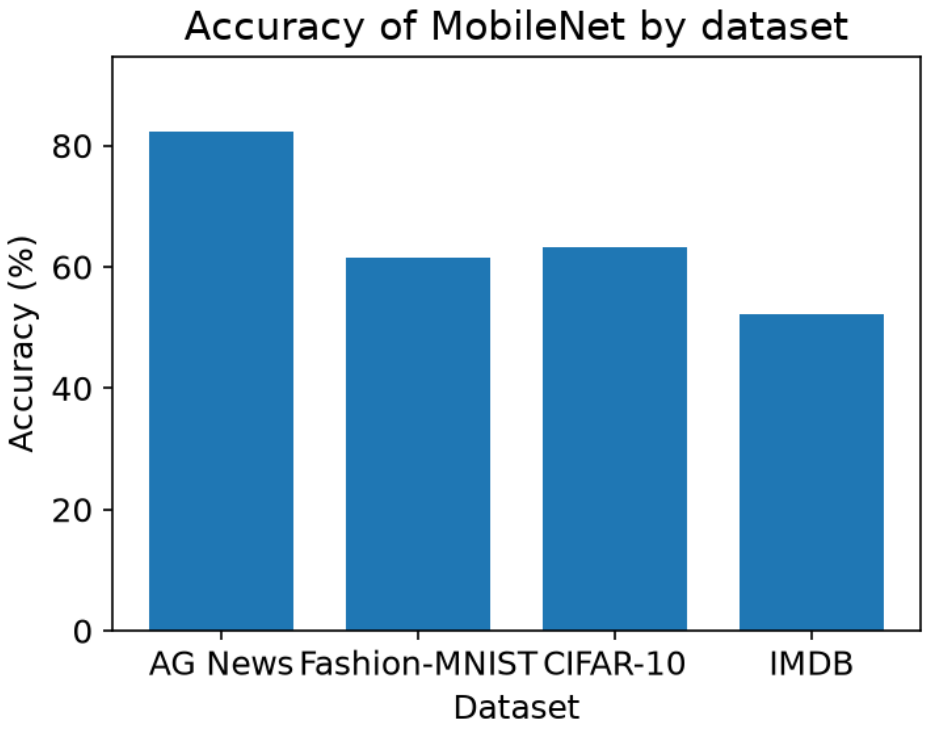


Figure 2: Accuracy of MobileNet by dataset for the models in this experiment.

As shown in Figure 2, MobileNet achieves 52% accuracy on IMDB. As shown in Figure 2, on AG News, MobileNet obtains a accuracy of about 80%. MobileNet reaches about 60% accuracy on Fashion-MNIST (Figure 2). MobileNet achieves 72.8% accuracy on CIFAR-10.

Figure 3 shows the validation accuracy of MLP by dataset.

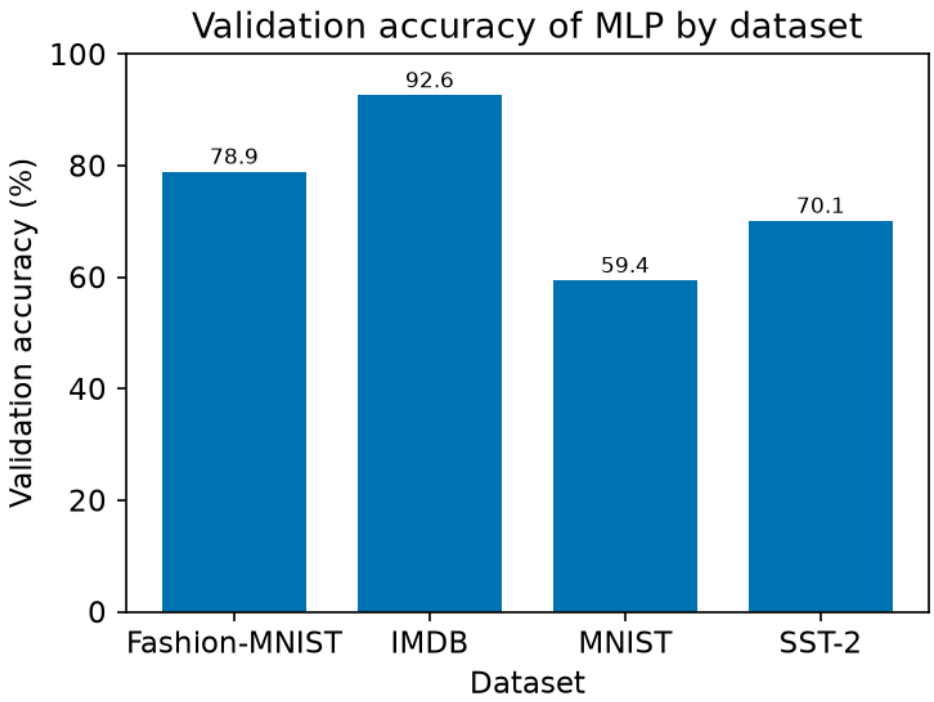


Figure 3: Validation accuracy of MLP by dataset for the models in this experiment.

As shown in Figure 3, MLP scores about 33 points higher on IMDB than on MNIST. As shown in Figure 3, on IMDB, MLP obtains a validation accuracy of 78.9%. As shown in Figure 3, on MNIST, MLP obtains a validation accuracy of 49.4%. As shown in Figure 3, MLP achieves 28.9% validation accuracy on Fashion-MNIST.

3 Discussion

The differences between the models are discussed in the next section.